

Code bài 1:

```
api_session_02 > app.py > ...
1 #app.py - bài 1: GET /books, POST /books
2 from flask import Flask, jsonify, request, make_response
3
4 app = Flask(__name__)
5 BOOKS = []
6 _next_id = 1
7
8 # - GET /books trả danh sách
9 @app.get("/books")
10 def list_books():
11     return jsonify({
12         "data": BOOKS,
13         "total": len(BOOKS)
14     }), 200
15
16 #- POST /books tạo mới
17 @app.post("/books")
18 def create_book():
19     global _next_id
20     if not request.is_json:
21         return jsonify(error="expected JSON"), 415
22     p = request.get_json(silent=True) or {}
23     t = (p.get("title") or "").strip()
24     a = (p.get("author") or "").strip()
25     if not t or not a:
26         return jsonify(error="title and author required"), 422
27     book = {"id": _next_id, "title": t, "author": a}
28     BOOKS.append(book); _next_id += 1
29     resp = make_response(jsonify(book), 201)
30     resp.headers["Location"] = f"/books/{book['id']}"
31     return resp
32
33 if __name__ == "__main__":
34     app.run()
```

1. GET danh sách rỗng

```
THI@LAPTOP-HCT3QG47 MINGW64 /e/SOA (main)
$ curl -i http://localhost:5000/books
HTTP/1.1 200 OK
Server: Werkzeug/3.1.8 Python/3.12.3
Date: Fri, 18 Sep 2026 14:37:45 GMT
Content-Type: application/json
Content-Length: 22
Connection: close

{"data": [], "total": 0}
```

2. POST tạo sách thành công

```

THI@LAPTOP-HCT3QG47 MINGW64 /e/SOA (main)
$ curl -i -X POST http://localhost:5000/books \
-H "Content-Type: application/json" \
-d '{"title":"Clean Code","author":"Robert Martin"}'
HTTP/1.1 201 CREATED
Server: Werkzeug/3.1.8 Python/3.12.3
Date: Fri, 18 Sep 2026 14:39:51 GMT
Content-Type: application/json
Content-Length: 55
Location: /books/1
Connection: close

{"author":"Robert Martin","id":1,"title":"Clean Code"}

```

3. POST thiếu title

```

THI@LAPTOP-HCT3QG47 MINGW64 /e/SOA (main)
$ curl -i -X POST http://localhost:5000/books \
-H "Content-Type: application/json" \
-d '{"author":"Robert Martin"}'
HTTP/1.1 422 UNPROCESSABLE ENTITY
Server: Werkzeug/3.1.8 Python/3.12.3
Date: Fri, 18 Sep 2026 14:41:15 GMT
Content-Type: application/json
Content-Length: 38
Connection: close

{"error":"title and author required"}

```

4. POST thiếu author

```

THI@LAPTOP-HCT3QG47 MINGW64 /e/SOA (main)
$ curl -i -X POST http://localhost:5000/books \
-H "Content-Type: application/json" \
-d '{"title":"Clean Code"}'
HTTP/1.1 422 UNPROCESSABLE ENTITY
Server: Werkzeug/3.1.8 Python/3.12.3
Date: Fri, 18 Sep 2026 14:42:16 GMT
Content-Type: application/json
Content-Length: 38
Connection: close

{"error":"title and author required"}

```

5. POST thiếu Content-Type

```

THI@LAPTOP-HCT3QG47 MINGW64 /e/SOA (main)
$ curl -i -X POST http://localhost:5000/books \
-d '{"title":"Clean Code","author":"Robert Martin"}'
HTTP/1.1 415 UNSUPPORTED MEDIA TYPE
Server: Werkzeug/3.1.8 Python/3.12.3
Date: Fri, 18 Sep 2026 14:43:02 GMT
Content-Type: application/json
Content-Length: 26
Connection: close

{"error":"expected JSON"}

```

6. POST JSON lỗi cú pháp

```
\
THI@LAPTOP-HCT3QG47 MINGW64 /e/SQA (main)
$ curl -i -X POST http://localhost:5000/books \
-H "Content-Type: application/json" \
-d '{"title":"Clean Code",}'
HTTP/1.1 422 UNPROCESSABLE ENTITY
Server: Werkzeug/3.1.8 Python/3.12.3
Date: Fri, 18 Sep 2026 14:43:35 GMT
Content-Type: application/json
Content-Length: 38
Connection: close

{"error":"title and author required"}
```